

# EV Charging Station Reservation System - Lab 1

## Requirements and Use-Case Flow

### 1) Requirements Table

| Req ID | Type       | Description                                                                                                                       | Priority | Acceptance Criteria                                                                                                                     | Rationale (short)                           |
|--------|------------|-----------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| FR-001 | Functional | The system shall match the driver vehicle connector type with available station chargers and reserve a 30-minute charging window. | High     | <b>Pass:</b> Incompatible connector plugs hidden from driver.<br><br><b>Fail:</b> Two vehicles booked for the same socket concurrently. | Core reservation functionality.             |
| FR-002 | Functional | The system shall allow the EV Driver to view available charging sockets based on current location and time.                       | High     | <b>Pass:</b> Map displays only stations with unreserved compatible slots.                                                               | Enables drivers to find chargers on the go. |
| FR-003 | Functional | The system shall calculate dynamic power rates based on grid load prior to reservation confirmation.                              | Medium   | <b>Pass:</b> Rate shown matches the pricing tier for the current grid load level.                                                       | Ensures transparent and dynamic pricing.    |

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| FR-002 | Functional | The system shall allow the EV Driver to view available charging sockets based on current location and time.                       | High     | <b>Pass:</b> Map displays only stations with unreserved compatible slots.                                                                      | Enables drivers to find chargers on the go. |
| FR-004 | Functional | The system shall connect to a Payment Gateway to process the transaction for the reserved slot.                                   | High     | <b>Pass:</b> Reservation is confirmed if and only if payment is authorized.                                                                    | Secures booking with financial commitment.  |
| FR-005 | Functional | The system shall allow the Station Operator to monitor the real-time status (locked/available/i                                   | Low      | <b>Pass:</b> Dashboard reflects state changes within 5 seconds of occurrence.                                                                  | Required for administrative monitoring.     |

| Req ID  | Type       | Description                                                                                                                               | Priority | Acceptance Criteria                                                                                                                            | Rationale (short)                            |
|---------|------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| FR-001  | Functional | The system shall match the driver vehicle connector type with available station chargers and reserve a 30-minute charging window.         | High     | <p><b>Pass:</b> Incompatible connector plugs hidden from driver.</p> <p><b>Fail:</b> Two vehicles booked for the same socket concurrently.</p> | Core reservation functionality.              |
| FR-002  | Functional | The system shall allow the EV Driver to view available charging sockets based on current location and time.                               | High     | <b>Pass:</b> Map displays only stations with unreserved compatible slots.                                                                      | Enables drivers to find chargers on the go.  |
|         |            | n-use) of all sockets.                                                                                                                    |          |                                                                                                                                                |                                              |
| NFR-001 | Nonfunc.   | The system shall automatically release locked sockets and apply a penalty if the driver fails to plug in within 10 minutes of slot start. | High     | <b>Pass:</b> Benchmarking tests confirm target latency and security standards under simulated peak load.                                       | Maximizes resource utilization during peaks. |

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| FR-002  | Functional | The system shall allow the EV Driver to view available charging sockets based on current location and time.                       | High     | <b>Pass:</b> Map displays only stations with unreserved compatible slots.                                                                      | Enables drivers to find chargers on the go. |
| NFR-002 | Nonfunc.   | The system shall respond to user search queries for charging stations within 2 seconds.                                           | Medium   | <b>Pass:</b> System logs show 95% of queries resolve under 2s under typical load.                                                              | Good usability and fast UX.                 |

## 2) Use-Case Flow Document

Use Case: Reserve Time Slot

Actors: EV Driver, Payment Gateway, Station operator

Pre-conditions: The EV Driver is logged in, has a valid payment method saved, and has located a compatible charging socket.

Post-conditions: The chosen charging socket is locked for the driver's specific 30-minute window, and a receipt is generated.

Main Success Scenario:

1. EV Driver selects an available 30-minute charging window for a compatible socket.
2. The system calculates the dynamic power rate and displays the total estimated cost.
3. EV Driver confirms the reservation details and proceeds to payment.
4. System requests payment method selection.
5. EV Driver selects a preferred payment method.
6. The system connects to the Payment Gateway (Include: Process Payment).
7. Payment Gateway authorizes the payment.
8. The system updates the backend, locks the socket for the designated window, and displays a confirmation message with reservation details to the EV Driver.
9. Use case ends successfully.

Alternate Flow:

7a. Payment Declined

- 7a1. The system displays a "Payment Failed" message.
- 7a2. EV Driver is prompted to select an alternative payment method.
- 7a3. If the payment fails again after 2 total attempts, the reservation process is cancelled, the time slot is unlocked, and the EV Driver is notified.

8a. Driver No-Show(Penalty Applied)

- 8a1. The reserved 30-minute charging window begins.
- 8a2. The system monitors the socket status for a vehicle connection.
- 8a3. If the EV Driver fails to plug in their vehicle within 10 minutes of the start time, the system automatically releases the locked socket.
- 8a4. The system applies a predefined financial penalty to the EV Driver's saved payment method.
- 8a5. The system notifies the EV Driver of the cancellation and the penalty charge.